

FINAL PROJECT · DELOITTE × LUISS 2026

DiscoverAI

Semantic Product Search for Health & Personal Care

A review-aware search system on Amazon Reviews 2023, with quality-driven re-ranking, BART summarization, and domain entity extraction.

Team Mean-Squared-Terrors

| May 3, 2026

THE CHALLENGE

Keyword search doesn't understand what shoppers actually mean

WHAT THE USER TYPES

Natural-language intent
"something to help me sleep at night"

Implicit exclusions
"face cream without parabens"

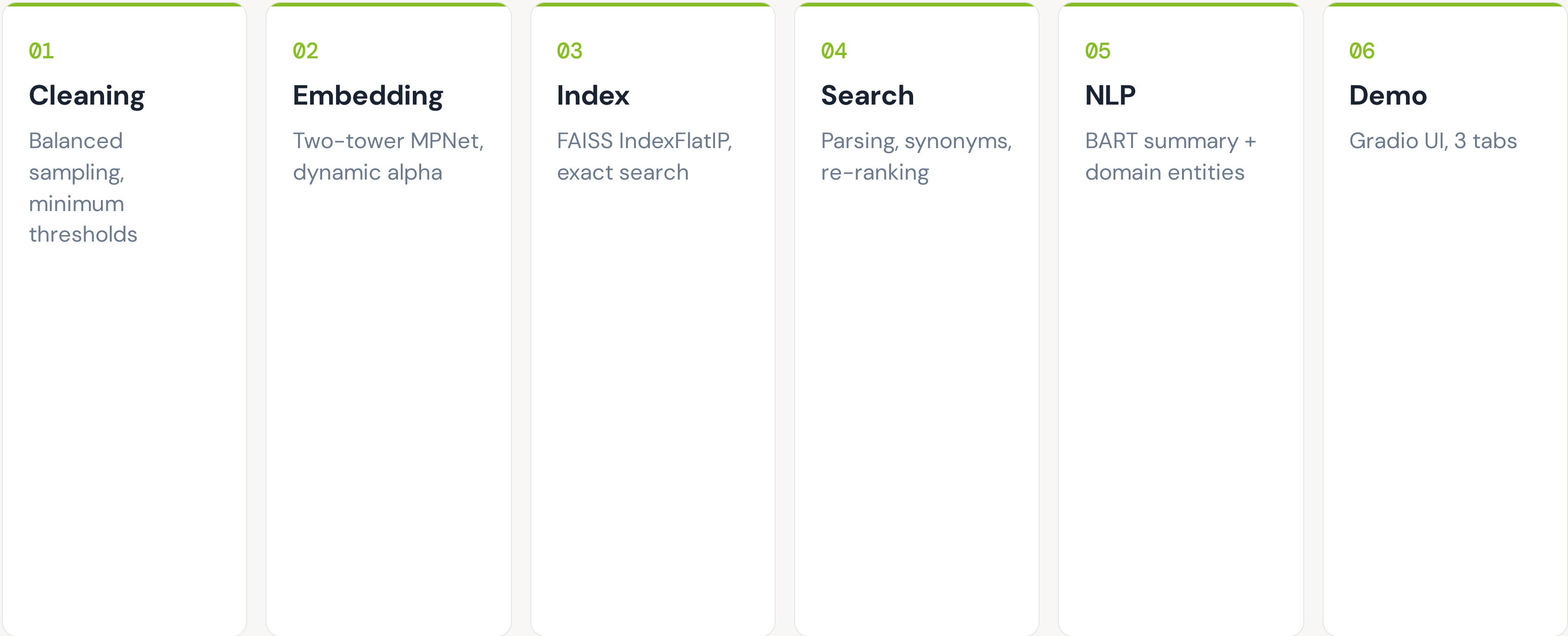
Price intent & trust signals
"professional but affordable shampoo"

WHAT TRADITIONAL SEARCH DOES

- Lexical match: looks for exact words only
- Doesn't understand synonyms or intent
- Cannot tell "with" from "without"
- Ranks by text relevance or raw popularity
- Ignores the customer voice inside reviews

A review-aware system that understands intent and trust

One embedding model, two sources of truth: the product text for what it claims to be, the reviews for how it actually performs, fused into a semantic index with quality-driven re-ranking.



DATASET AT A GLANCE

Amazon Reviews 2023: Health & Personal Care

Field coverage, token distribution and reviews per product (balanced set)

7,647

products

318k

cleaned reviews

57k

balanced reviews

96%

brand coverage

CLEANING DECISIONS

- Min 10 reviews per product
Below this threshold the review signal is too weak for a meaningful embedding
- Drop reviews shorter than 25 tokens
"Love it!!", "Fast shipping" carry no semantic content

Product field coverage (%)

Price	26%
Details/Specs	57%
Description	31%
Features	35%
Brand	96%

Product-level overview

Tokens per product (product_text_base)

Reviews per product (balanced)

REDUCING THE 5-STAR BIAS

Balanced sampling: from 61.7% to 46.2% five-star reviews



WHY THIS MATTERS

Without balancing, the model sees a corpus where 6 out of 10 reviews say "5 stars, love it" — useful signal drowns in praise. The bias correction is per-product, so even niche items keep a critical voice.

HOW THE CAP WORKS

{5:5, 4:3, 3:2, 2:3, 1:3}

Per product, take at most this many reviews per rating, prioritising the most helpful within each band.

Two-tower embedding: product and reviews kept separate

Two sources, two voices

Product text says what the brand sells; reviews say how it actually works. Concatenating lets the noisier source dominate.

Same encoder, two embeddings

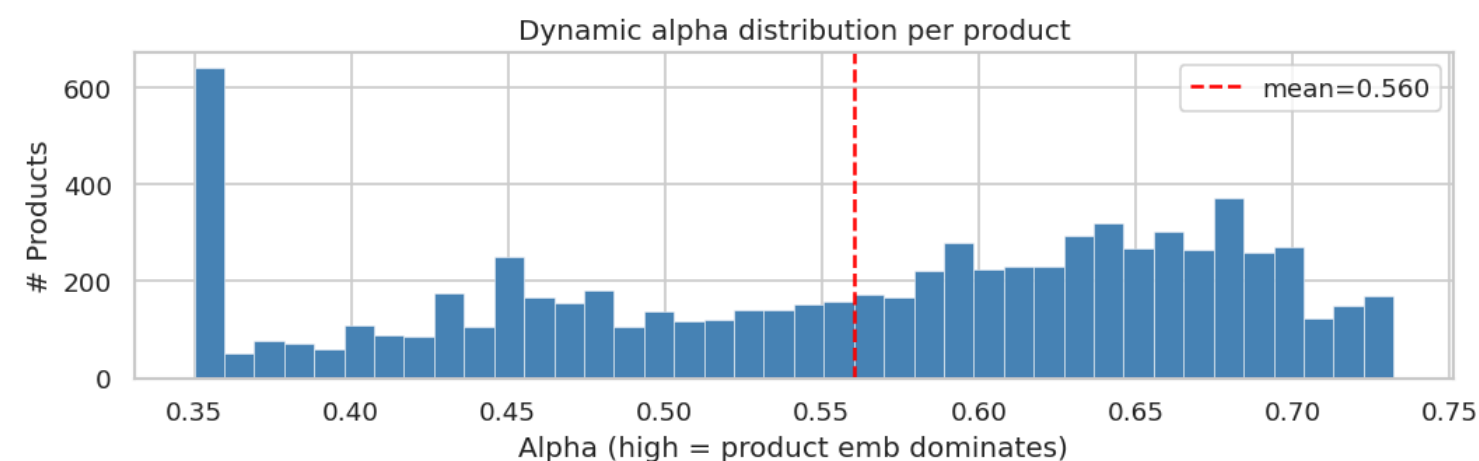
all-mpnet-base-v2 (768 dim, trained on 1B+ pairs): shared semantics, vectors we can weight separately.

Weighted review aggregation

70% helpful_vote score + 30% deviation from neutral rating: most useful reviews surface first.



Dynamic alpha: the product decides the weight, not the engineer



Real distribution of α across the 7,593 products (mean 0.560)

WEAK SIGNAL → HIGH A

Product text dominates

Few reviews, low helpful_vote → product text carries the representation.

$\alpha \approx 0.70$

STRONG SIGNAL → LOW A

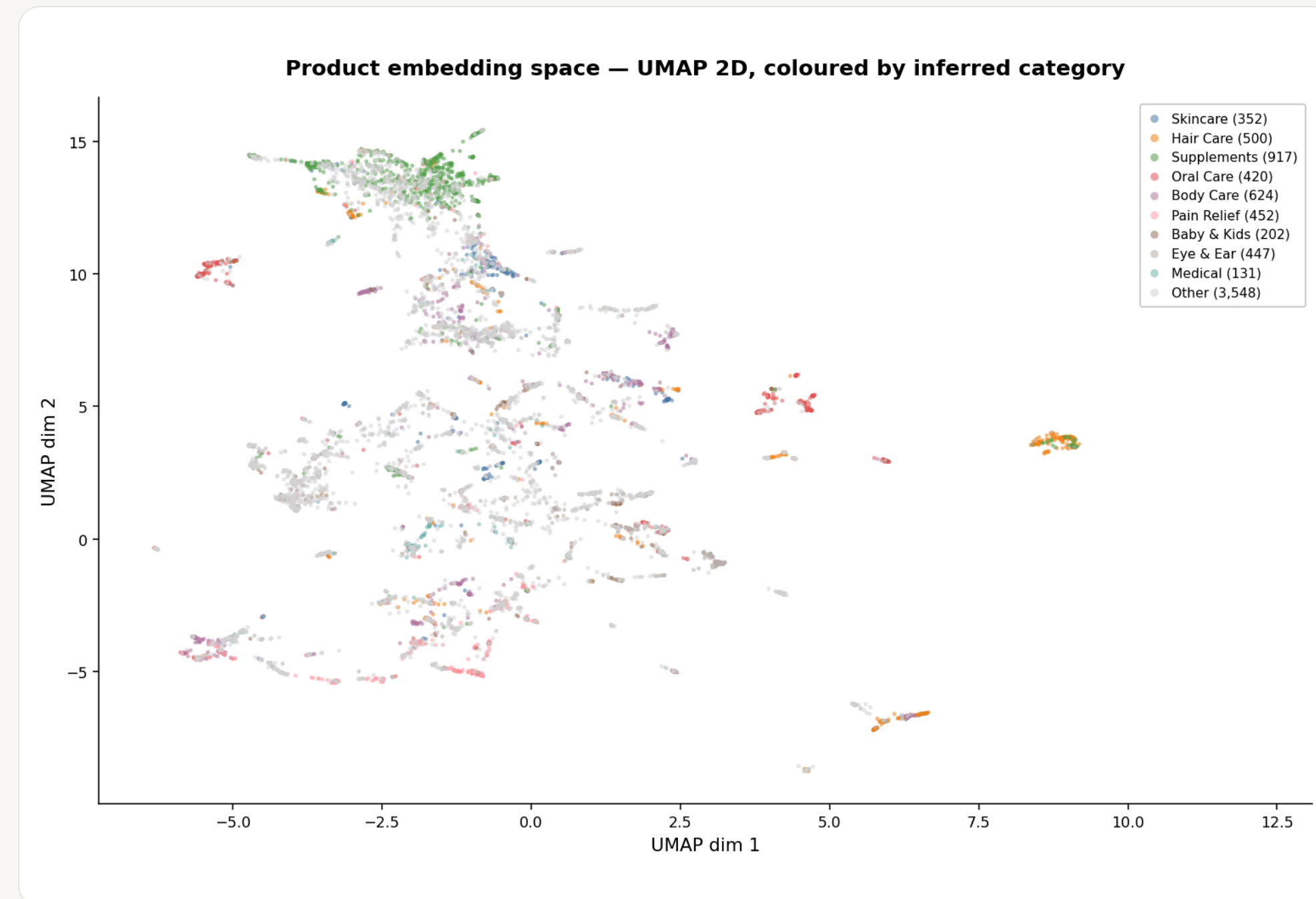
Customer voice carries weight

Many helpful reviews, decisive rating → customer voice carries weight.

$\alpha \approx 0.35$

UMAP projection: semantic clusters emerge by category

7,593 product embeddings · UMAP 2D · coloured by inferred category



WHAT IT TELLS US

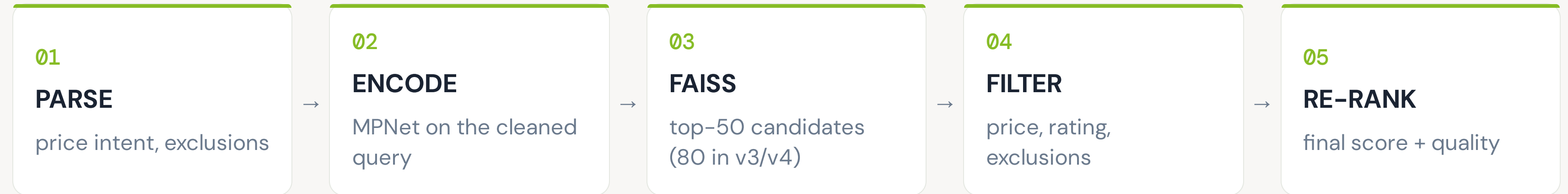
- Supplements form a tight cluster top-left.
- Hair Care separates cleanly from Skin Care.
- Pain Relief sits between supplements and topical products — it bridges both.
- The model learned the catalogue's structure without ever being trained on it: proof of useful semantic geometry.

FAISS IndexFlatIP: exact search, no compromises

OPTION	LATENCY @ 7.6K	RECALL	TUNING
IndexFlatIP ✓	negligible	1.0	none
IndexIVF	lower	≈ 0.95	nlist, nprobe to tune
IndexHNSW	lower	≈ 0.97	M, efSearch, slow build

Defense: at 7,593 products exact search is effectively instant. An approximate index trades recall for speed we wouldn't notice. Approximate becomes the right answer once the catalogue grows past **tens of millions**.

From query to ranked results



FINAL SCORE

$$\text{score} = \text{similarity} + \beta_{\text{quality}} \cdot \text{quality_score} + \beta_{\text{pop}} \cdot \text{popularity_score}$$

$\beta_{\text{quality}} = 0.12 \cdot \beta_{\text{pop}} = 0.00$ (disabled after ablation) $\rightarrow$ small enough that a popular but irrelevant product cannot leapfrog a better match

QUALITY SCORE

$$0.40 \cdot \text{rating_norm} + 0.35 \cdot \text{sentiment} + 0.25 \cdot \text{helpful_credibility}$$

popularity_score: log-normalised review_count, clipped at the 95th percentile to avoid outlier dominance

Guardrail: blocking off-topic queries before they fail silently

Without it, "who won the Champions League?" returns supplements with low similarity — silently bad answers instead of an explicit rejection.

L1 · $O(1)$

Structural check

Length bounds (2–60 tokens) and a regex blacklist on clearly off-topic terms (sports, weather, politics, recipes, tech). Positive in-domain signals (shampoo, retinol, probiotic...) short-circuit the next layers.

L2 · $O(d)$

Centroid similarity

One dot product against the corpus centroid. Catches queries that lie far from the H&PC distribution but escape the blacklist.

L3 · $O(\log n)$

FAISS top-1 confidence

Cosine of the closest product. Below 0.18 the query has no real match in the catalogue → reject with explicit message instead of a noisy result.

TEST RESULT

F1 = 0.957

on 24 queries (12 in-domain, 12 off-topic) ·
precision 1.0 · recall 0.917

EXAMPLES

PASS "vitamin D supplement 1000 IU" — top-1 sim 0.78

PASS "shampoo for dry hair without sulfates" — top-1 sim 0.71

BLOCK "who won the Champions League?" — L1 blacklist

BLOCK "best laptop for gaming" — L1 blacklist

Iterating through versions: Demo now uses hybrid_v4

v1 · BASELINE

Similarity + fixed re-ranking

The foundation

- FAISS top-50 + score with fixed β
- Works on "catalogue-style" queries
- Fails on negation and natural language

v2 · NEGATION

Filter out what users exclude

Answers observed failure

- Patterns: "without X", "no X", "X-free"
- Extracted terms post-filter the product text
- "cream without parabens" stops returning paraben creams

v3 · PRODUCTION

Synonyms, price, dosages, adaptive β

Full-featured

- Synonym expansion ("SPF" → sunscreen...)
- Price-intent parsing ("affordable" → budget)
- Dosage extraction (1000mg, SPF 50...)
- Adaptive β_{quality} : 0.20 when mean sim < 0.65

v4 · PRODUCTION (best version)

Hybrid v4: BM25 + Semantic + RRF

Powers the demo

- Demo powered by hybrid_v4
- BM25 + Semantic + Reciprocal Rank Fusion
- Evaluation — NDCG@5 = 0.526

5 systems · 37 queries · pseudo-relevance judgments

Hand-crafted benchmark with 4-level relevance labels (keyword matching on product_text_base). Caveat: this method mildly favours BM25.

SYSTEM	NDCG@5	MRR	RECALL@5	P@5	NOTES
bm25	0.535	0.811	0.153	0.632	strong on lexical / brand
semantic	0.526	0.855	0.138	0.573	best top-1
search_base	0.517	0.780	0.135	0.568	semantic + quality re-rank

NDCG@5 BY INTENT — WINNER PER ROW		
intent	winner	score
brand	bm25	0.647
dosage	bm25	0.906
price	bm25	0.438
negation	search_v3	0.498
semantic	hybrid_v4	0.581

READING

BM25 wins on lexical specifics (brand, dosage, price); **hybrid_v4** wins on semantic queries and matches BM25 on NDCG@5 overall — the best of both at the cost of a small MRR drop vs pure semantic.

SUMMARIZATION & EXTRACTION

From thousands of reviews to a readable card in seconds

SUMMARIZATION

BART-Large-CNN

- Split into positive (≥ 4 ★) and negative (≤ 2 ★)
- Filters: ≥ 12 words, no "I...", < 300 chars, deduped
- BART produces the abstractive summary; pros/cons are extracted from positive/negative buckets
- Extractive fallback if < 5 reviews — avoids hallucinations on thin evidence

ENTITY RECOGNITION

Domain regex + spaCy NER

ACTIVE INGREDIENTS

retinol niacinamide hyaluronic acid SPF omega-3

CERTIFICATIONS

organic vegan cruelty-free paraben-free

USE-CASE TARGETS

skin type hair type age group body area

Gradio interface, 3 tabs, the entire pipeline integrated

01

Search

Free-text query with optional filters for price bucket and minimum rating.

- product image, rating, price
- BART summary
- extracted pros & cons

02

Similar Products

Content-based recommendation: from a product name, returns the 5 most similar on the combined embedding.

- no text encoding at runtime
- operates directly on combined_emb

03

System Info

System transparency for evaluators: index size, model in use, feature coverage.

- index stats
- model details
- feature coverage

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Thank You

Questions